

Notes: Huang, Mayer, and Miller (RFS, 2024)
Gender Bias in Promotions: Evidence from Financial Institutions

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1 Big picture

This paper studies whether women face gender bias at the first major promotion step inside financial institutions. The setting is U.S. mortgage companies, also called shadow banks, where mortgage loan officers can be promoted to mortgage managers. The authors build a nationwide worker-manager panel for roughly 72,000 workers at more than 1,000 firms from 2014 to 2019.

The paper asks a difficult question: does a gender promotion gap reflect discrimination, or could it reflect unobserved differences in preferences, experience, or managerial potential? The key contribution is to move beyond standard promotion-gap regressions and implement Becker’s outcome-test logic at the margin of promotion. If firms are unbiased, marginally promoted men and women should have the same expected performance as managers. If marginally promoted women outperform marginally promoted men, then firms are holding women to a higher promotion standard.

The central result is that female loan officers are less likely to be promoted than comparable male loan officers, and marginally promoted female managers subsequently perform better than marginally promoted male managers. This combination is interpreted as evidence of gender bias in promotions.

2 Incremental contribution of the paper

1. **New data on rank-and-file finance workers.** The paper constructs a national panel of loan officers and their managers at mortgage companies using NMLS licensing records, then links workers to loan-level production data and later foreclosure outcomes.
2. **Direct test of bias rather than only disparity.** The paper documents a gender promotion gap, but its main advance is using Becker’s marginal outcome test to distinguish gender bias from unobserved worker heterogeneity.
3. **Instrumental-variable implementation of the outcome test.** The authors use leave-out firm-year promotion rates as an instrument for promotion, identifying marginally promoted workers whose promotion occurs because the firm promotes unusually many workers that year.
4. **Mechanism evidence through in-group tests.** The paper tests whether bias is smaller under female managers. Similar bias under male and female managers points more toward inaccurate stereotypes than male-only taste-based animus.
5. **Firm-cost evidence.** The paper connects biased promotion practices to firm outcomes, showing that firms with larger gender promotion gaps have lower loan volume, smaller employment, and weaker survival patterns.

3 Data and measurement

3.1 Core data sources

- **NMLS licensing records.** These records identify loan officers, employers, branches, tenure, and manager-subordinate links at mortgage companies. The data allow the authors to observe promotions to mortgage manager.

- **CoreLogic loan-level data.** Loan officer NMLS IDs are linked to originated mortgages, allowing objective measures of worker output: number of loans and dollar loan volume.
- **Zillow ZTRAX foreclosure data.** Foreclosure outcomes are linked to originated loans, allowing robustness tests on loan quality.
- **Gender and demographics.** Gender is inferred from names and extensive manual web searches; race/ethnicity is assigned using Bayesian Improved First Name Surname Geocoding.

3.2 Key variables

- **Promoted:** indicator for whether a loan officer becomes a manager at the same firm in the next year.
- **Externally promoted:** indicator for whether a loan officer becomes a manager at a different mortgage company.
- **Managerial effect:** manager value added, measured as the manager fixed effect in subordinate loan volume regressions.
- **Female:** indicator for a female loan officer or manager.
- **Gender gap in promotions:** male promotion rate minus female promotion rate at the firm-year level.

4 Economic model and empirical logic

4.1 Becker outcome-test logic

The paper is built around two predictions from Becker’s model of discrimination:

1. **Higher bar prediction.** If firms are biased against women, then marginally promoted women must have higher managerial potential than marginally promoted men.
2. **Cost prediction.** If firms forgo efficient promotion decisions because of bias, biased firms should incur economic costs.

The key challenge is the infra-marginality problem. Comparing average performance of all promoted men and women can be misleading because promoted workers far above the promotion threshold are infra-marginal. The relevant comparison is among workers at the promotion margin.

5 Empirical specifications

5.1 Gender promotion gap

The baseline promotion-gap regression is:

$$Promoted_{i,t} = \beta_1 Female_i + \beta_2 X_{i,t} + \eta_{j,t} + \eta_k + \varepsilon_{i,t},$$

where i indexes workers, t years, j firms, and k branches. The controls include performance, tenure, race/ethnicity, firm-by-year fixed effects, branch fixed effects, and volume-decile-by-year fixed effects.

5.2 Managerial performance

Managerial effect is estimated from subordinate performance regressions:

$$LoanVolume_{i,m,j,t} = \beta_1 X_{i,t} + \eta_i + \eta_m + \eta_{j,t} + \varepsilon_{i,t}.$$

The manager fixed effect η_m is the paper’s measure of manager performance. Loan officer fixed effects ensure that managers are evaluated based on changes in the same subordinate’s productivity under different managers, rather than on the unconditional quality of their teams.

5.3 Instrumental-variable outcome test

The authors instrument for promotion using the leave-out firm-year promotion rate, excluding the focal worker and branch peers:

$$Promoted_{i,t} = \beta_0 + \beta_1 LeaveOutPromotionRate_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t},$$

$$ManagerialEffect_{i,t} = \beta_0 + \beta_1 \widehat{Promoted}_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t}.$$

The IV identifies the local average treatment effect for compliers: workers promoted because their firm has a high promotion-rate year. These are interpreted as marginally promoted workers, exactly the group required for Becker’s outcome test.

6 Main findings

- Women represent about 35% of loan officers but only about 27% to 29% of managers during the sample period.
- Female loan officers are about 0.17 percentage points less likely to receive an internal promotion each year after controls, a 15% decline relative to the base internal promotion rate.
- Female loan officers are about 0.44 percentage points less likely to receive external promotions, a 29% decline relative to the base external-promotion rate.
- Average managerial performance does not significantly differ by gender, illustrating why naive outcome tests are weak.
- IV outcome tests show that marginally promoted women outperform marginally promoted men by about 6% in manager value added, implying that women face a higher promotion bar.
- In-group tests show bias under both male and female managers, consistent with gender stereotypes rather than only male taste-based discrimination.

- Robustness tests show that the higher measured performance of marginal female managers is not offset by worse loan quality or weaker workforce management.
- Firm-level evidence is consistent with Becker's cost prediction: firms with larger gender promotion gaps have lower loan volume, smaller employment, and weaker survival patterns.

7 Tables and figures

7.1 Figure 1. The importance of mortgage companies in the U.S. mortgage market and Figure 2. Illustration of the infra-marginality problem

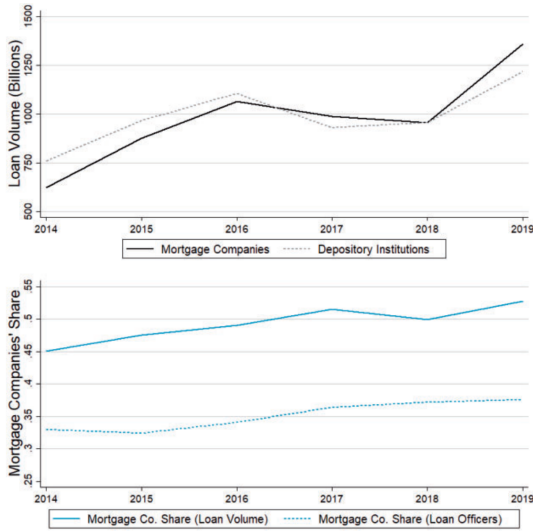


Figure 1
The importance of mortgage companies in the U.S. mortgage market
This figure shows the importance of mortgage companies in the U.S. mortgage market during our sample period of 2014 to 2019. The top plot presents the total mortgage origination volume across all loan types by mortgage companies (black line) and depository institutions (dashed line). The bottom plot presents mortgage companies' share of the market based on loan volume (blue line) and number of loan officers (dashed line).

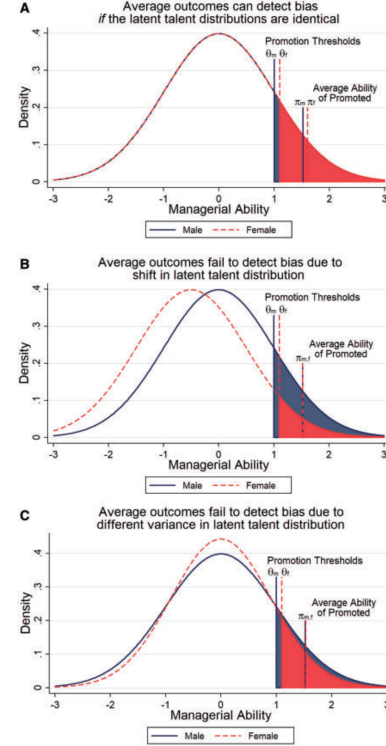


Figure 2
Illustration of the infra-marginality problem
This figure illustrates the infra-marginality problem in our context through three examples. The examples each plot hypothetical distributions of male and female loan officers' managerial ability, draw promotion thresholds (θ) above which loan officers are promoted to manager, and consider the average ability (π) of promoted managers. Male loan officers' managerial ability is distributed standard normal in each example, whereas the female distribution varies. The examples all include gender bias in promotion decisions ($\theta_f = 1.1$, whereas $\theta_m = 1$). Panel A shows that if female managerial ability follows an identical standard normal distribution as its male counterpart, then gender bias can be detected in average outcomes since $\pi_f > \pi_m$. Panel B shows that if the female distribution of managerial ability is shifted left, then the gender bias can be obscured in average outcomes (note that $\pi_f = \pi_m$). Panel C shows that even if the female distribution is centered at zero, but is narrower, then the bias is again obscured in average outcomes. See Section 4.2 for additional discussion.

Figure 1. The importance of mortgage companies in the U.S. mortgage market

Figure 2. Illustration of the infra-marginality problem

Read:

- Figure 1 shows that mortgage companies became central to U.S. mortgage origination during the sample period, with loan volume exceeding that of depository institutions by 2019.
- Figure 2 explains why average outcomes can fail to detect discrimination: the average promoted female manager can look similar to the average promoted male manager even if the female promotion cutoff is higher.

Economic message: The setting is economically important and provides enough microdata to study early-career promotions. The key methodological point is that the relevant test of bias is not whether average promoted women outperform average promoted men, but whether marginally promoted women outperform marginally promoted men.

7.2 Table 1. Descriptive statistics and Table 2. The gender gap in promotions

Table 1

Descriptive statistics

A. Female representation

	(1) Total number of loan officers	(2) % with gender identified	(3) % female	(4) Total number of managers	(5) % with gender identified	(6) % female
2014	20,634	99.32	34.40	4,380	99.89	26.33
2015	25,995	99.40	34.17	5,189	99.90	26.37
2016	29,833	99.48	34.04	5,818	99.90	27.06
2017	32,587	99.45	34.44	6,378	99.91	27.51
2018	32,615	99.45	35.60	6,594	99.89	28.37
2019	33,669	99.25	36.38	6,879	99.93	28.69

B. Summary statistics for loan officers

	(1) All LOs (N=174,272)	(2) Mean	(3) SD	(4) Female LOs (N=60,847)	(5) Mean	(6) SD	(7) p-value
<i>Female</i>		0.3491	0.4767	–	–	–	–
<i>Promoted</i>		0.0115	0.1066	0.0114	0.1063	0.0115	0.1068 .8121
<i>Firm-year promotion rate</i>		0.0115	0.0192	0.0123	0.0194	0.0111	0.0191 .0000
<i>Externally promoted</i>		0.0150	0.1214	0.0131	0.1138	0.0160	0.1253 .0000
<i>Number of loans</i>		36.5262	47.9252	30.9753	41.0913	39.5039	50.9684 .0000
<i>log(number of loans)</i>		2.8910	1.2801	2.7687	1.2333	2.9567	1.2998 .0000
<i>Loan volume (million \$)</i>		8.5121	11.4459	7.1206	9.8253	9.2585	12.1616 .0000
<i>log(loan volume)</i>		15.0676	1.4974	14.9120	1.4610	15.1510	1.5100 .0000
<i>Foreclosure ratio</i>		0.0079	0.0293	0.0081	0.0300	0.0077	0.0289 .0060
<i>Foreclosure amount</i>		0.0009	0.0053	0.0008	0.0052	0.0009	0.0053 .0036
<i>White</i>		0.7422	0.4374	0.7276	0.4452	0.7501	0.4330 .0000
<i>Tenure</i>		2.3245	1.8195	2.4896	1.9139	2.2360	1.7604 .0000

C. Summary statistics for promoted loan officers (year immediately before promotion)

	(1) All LOs (N=2,005)	(2) Mean	(3) SD	(4) Female LOs (N=695)	(5) Mean	(6) SD	(7) p-value
<i>Number of loans</i>		38.9107	45.3461	38.5727	43.6473	39.0901	46.2374 .8080
<i>log(number of loans)</i>		3.0712	1.1997	3.1008	1.1621	3.0554	1.2193 .4207
<i>Loan volume (million \$)</i>		9.2887	11.4235	8.7848	10.4327	9.5560	11.9110 .1503
<i>log(loan volume)</i>		15.2839	1.4051	15.2798	1.3565	15.2860	1.4308 .9247
<i>Foreclosure ratio</i>		0.0127	0.0376	0.0120	0.0350	0.0130	0.0389 .5881
<i>Foreclosure amount</i>		0.0011	0.0057	0.0009	0.0053	0.0011	0.0060 .4452
<i>White</i>		0.8030	0.3978	0.7856	0.4107	0.8122	0.3907 .1542
<i>Tenure</i>		2.3127	1.7808	2.4446	1.8431	2.2427	1.7434 .0157

This table presents descriptive statistics for our sample. Panel A reports female representation among loan officers and managers. Column 1 shows the total number of loan officers in our sample each year, column 2 shows the percentage with gender identified, and column 3 shows the percentage of female officers. Columns 4–6 present the same statistics for managers in our sample. Panels B and C report summary statistics for our loan officer-year panel and the promoted loan officer sample, respectively. Columns 1 and 2 report the mean and standard deviation (SD) for the full sample. Columns 3 and 4 report the mean and standard deviation (SD) for the female sample. Columns 5 and 6 report the mean and standard deviation (SD) for the male sample. Column 7 reports the *p*-value of the difference between the female and male subsamples. Table A.1 in the appendix defines all variables.

Table 1. Descriptive statistics

Table 2

The gender gap in promotions

	(1)	(2)	(3)	(4)
<i>A. Internal promotions</i>				
Dependent variable:		<i>Promoted</i>		
<i>Female</i>	–0.0014** (0.0006)	–0.0021*** (0.0006)	–0.0017*** (0.0006)	–0.0017*** (0.0006)
<i>Tenure</i>			0.0002 (0.0002)	0.0002 (0.0002)
<i>White</i>			0.0014*** (0.0006)	0.0014*** (0.0006)
<i>log(number of loans)</i>			0.0049*** (0.0003)	
Firm x year FE	Y	Y	Y	Y
Branch FE	N	Y	Y	Y
Volume decile x year FE	N	N	N	Y
Observations	174,015	171,182	171,182	171,182
R-squared	.0283	.1397	.1421	.1428
<i>B. External promotions</i>				
Dependent variable:		<i>Externally promoted</i>		
<i>Female</i>	–0.0041*** (0.0006)	–0.0047*** (0.0007)	–0.0043*** (0.0007)	–0.0044*** (0.0007)
<i>Tenure</i>			–0.0016*** (0.0002)	–0.0016*** (0.0002)
<i>White</i>			0.0004 (0.0007)	0.0005 (0.0007)
<i>log(number of loans)</i>			0.0039*** (0.0003)	
Firm x year FE	Y	Y	Y	Y
Branch FE	N	Y	Y	Y
Volume decile x year FE	N	N	N	Y
Observations	174,015	171,182	171,182	171,182
R-squared	.0347	.1443	.1454	.1460

This table reports OLS regressions testing whether loan officers' gender affects the likelihood of being promoted to manager. The unit of observation is at the loan officer-year level. The dependent variables in panels A and B are indicators for the loan officer being promoted in the next year within the firm and outside of the firm, respectively. The key independent variable is the *Female* indicator. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. **p* < .1; ***p* < .05; ****p* < .01.

Table 2. The gender gap in promotions

Read:

- Table 1 documents the sample: female representation is higher among loan officers than among managers; women are about 35% of loan officers but less than 30% of managers.
- Table 2 shows a statistically significant gender promotion gap. The internal promotion coefficient is about –0.0017 in the most saturated specification, and the external promotion gap is about –0.0044.

Economic message: Women are underrepresented in management, and the gap remains after comparing workers in the same firm-year and branch and after controlling for output, tenure, and race. This establishes the promotion disparity that motivates the bias tests.

7.3 Table 3. Average managerial performance by gender and Table 4. Gender bias in promotions

Table 3

Average managerial performance by gender

	(1)	(2)	(3)	(4)
	<i>Managerial effect on number of loans</i>		<i>Managerial effect on loan volume</i>	
<i>A. Summary statistics for newly promoted managers' managerial effect</i>				
	Mean	SD	Mean	SD
All (N = 2,005)	−0.0089	0.1582	−0.0128	0.1866
Female (N=695)	−0.0041	0.1543	−0.0089	0.1709
Male (N=1,310)	−0.0115	0.1602	−0.0148	0.1945
<i>B. Regression results</i>				
Female	0.0074 (0.0074)	0.0077 (0.0072)	0.0059 (0.0085)	0.0067 (0.0084)
Controls	N	Y	N	Y
Observations	2,005	2,005	2,005	2,005
R-squared	.0005	.0226	.0002	.0212

This table presents statistics and tests examining gender differences in newly promoted managers' average performance. Panel A reports summary statistics on *Managerial effect* for newly promoted managers. Columns 1 and 2 report the mean and standard deviation (SD) of *Managerial effect on number of loans*. Columns 3 and 4 report the mean and standard deviation (SD) of *Managerial effect on loan volume*. Panel B presents OLS regression results testing whether *Managerial effect* differs by gender among the newly promoted managers. The dependent variables in columns 1 and 2 and columns 3 and 4 are the managerial effect on subordinates' number of loans, and on total loan volume, respectively. The key independent variable is the *Female* indicator. The controls include *Tenure*, *White*, and prepromotion volume decile bins interacted with year. Table A.1 in the appendix defines all variables. The standard errors are heteroskedasticity-robust. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4

Gender bias in promotions

	(1)	(2)	(3)	(4)
	Managerial effect on number of loans		Managerial effect on loan volume	
Dependent variable:	Male	Female	Male	Female
Subsample:				
Promoted	-0.0352** (0.0174)	0.0287** (0.0134)	-0.0458** (0.0209)	0.0332* (0.0173)
Controls	Y	Y	Y	Y
Observations	113,425	60,847	113,425	60,847
First-stage F-stat	48.8083	23.1582	48.8083	23.1582
p-value		.0036***		.0036***

This table presents IV regressions testing for differences in managerial performance between marginally promoted male and female managers. The unit of observation is at the loan officer-year level. The dependent variables in columns 1 and 2 and columns 3 and 4 are the managerial effect on subordinates' number of loans, and on total loan volume, respectively. We extract these managerial effects from the manager fixed effects estimation for each manager by regressing each subordinate's loan volume on loan officer fixed effects, manager fixed effects, firm-year fixed effects, and a control for loan officer tenure (see Equation (2)). The key independent variable, *Promoted*, is an indicator that equals one if the loan officer is promoted in the next year and zero otherwise. We instrument for *Promoted* with a leave-out measure of the average promotion rate of loan officers at the same firm in the same year, excluding the focal loan officer and other officers working at the same branch. The sample is split on gender in order to identify differences between the marginally promoted male manager (columns 1 and 3) and the marginally promoted female manager (columns 2 and 4). The final row reports p -values from a Wald test of equality at the margin, that is, between the *Promoted* coefficients in the male/female tests. The controls include *Tenure*, *White*, and volume decile bins interacted with year. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3. Average managerial performance by gender

Table 4. Gender bias in promotions

Read:

- Table 3 shows that average newly promoted female managers do not significantly outperform newly promoted male managers.
- Table 4 shows the marginal outcome-test result: marginally promoted male managers have negative value added, while marginally promoted female managers have positive value added. The difference is statistically significant.

Economic message: The average comparison hides the key result. Once the authors focus on marginal promotions using IV, women appear to face a higher promotion threshold. This is the central evidence of gender bias.

7.4 Table 5. In-group tests: previous manager gender and Table 6. Robustness test ruling out differences in loan quality

Table 5

In-group tests: The previous manager's gender and promotion equity

	(1)	(2)	(3)	(4)
<i>A. Managerial effect on number of loans</i>				
Subsample:	Male previous manager		Female previous manager	
	Male	Female	Male	Female
Promoted	−0.0403* (0.0216)	0.0000 (0.0001)	−0.0143 (0.0159)	0.0860* (0.0487)
Controls	Y	Y	Y	Y
Observations	95,725	45,240	17,630	15,536
First-stage F-stat	48.0458	11.3778	3.98072	9.3746
p-value		.0615*		.0506*
<i>B. Managerial effect on loan volume</i>				
Subsample:	Male previous manager		Female previous manager	
	Male	Female	Male	Female
Promoted	−0.0570** (0.0265)	−0.0000 (0.0001)	−0.0074 (0.0082)	0.1018* (0.0607)
Controls	Y	Y	Y	Y
Observations	95,725	45,240	17,630	15,536
First-stage F-stat	48.0458	11.3778	3.98072	9.3746
p-value		.0318**		.0746*

This table presents IV regressions testing for differences in managerial performance between marginally promoted male and female managers, split based on whether their previous manager (a likely participant in promotion decisions) was male versus female. The unit of observation is at the loan officer-year level. The dependent variables in panels A and B are the managerial effect on subordinates' number of loans, and on total loan volume, respectively. We extract these managerial effects from the manager fixed effects estimation for each manager by regressing each subordinate's loan volume on loan officer fixed effects, manager fixed effects, firm-year fixed effects, and a control for loan officer tenure (see Equation (2)). The key independent variable, *Promoted*, is an indicator that equals one if the loan officer is promoted in the next year and zero otherwise. We instrument for *Promoted* with a leave-out measure of the average promotion rate of loan officers at the same firm in the same year, excluding the focal loan officer and other officers working at the same branch. The sample is split first based on whether the previous manager was male (columns 1 and 2) or female (columns 3 and 4), and then based on loan officers' own gender to test for gender differences at the margin. The final row reports *p*-values from a Wald test of equality at the margin, that is, between the *Promoted* coefficients in the male/female loan officer tests. The controls include *Tenure*, *White*, and volume decile bins interacted with year. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. **p* < .1; ***p* < .05; ****p* < .01.

Table 6

Robustness test ruling out differences in loan quality under female managers

	(1)	(2)	(3)	(4)
<i>Managerial effect on foreclosure ratio</i>				
Subsample:	Male		Female	
	Male	Female	Male	Female
Promoted	−0.0138 (0.0090)	0.0005 (0.0030)	−0.0529 (0.0379)	0.0031 (0.0463)
Controls	Y	Y	Y	Y
Observations	113,425	60,847	113,425	60,847
First-stage F-stat	48.8083	23.1582	48.8083	23.1582
p-value		.1298		.3486

This table reports robustness tests based on loan quality measures. We use IV regressions to test for differences between the marginally promoted male versus female manager's impact on the loan quality of their subordinates. The unit of observation is at the loan officer-year level. The dependent variables in columns 1 and 2 and columns 3 and 4 are the managerial effect on subordinates' foreclosure ratio, and on their foreclosure amount, respectively. We extract these managerial effects from the manager fixed effects estimation for each manager by regressing each subordinate's foreclosure statistics on loan officer fixed effects, manager fixed effects, firm-year fixed effects, and a control for loan officer tenure (see Equation (2)). The key independent variable, *Promoted*, is an indicator that equals one if the loan officer is promoted in the next year and zero otherwise. We instrument for *Promoted* with a leave-out measure of the average promotion rate of loan officers at the same firm in the same year, excluding the focal loan officer and other officers working at the same branch. The sample is split on gender in order to identify differences between the marginally promoted male manager (columns 1 and 3) and the marginally promoted female manager (columns 2 and 4). The final row reports *p*-values from a Wald test of equality at the margin, that is, between the *Promoted* coefficients in the male/female tests. The controls include *Tenure*, *White*, and volume decile bins interacted with year. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. **p* < .1; ***p* < .05; ****p* < .01.

Table 5. In-group tests: previous manager gender and promotion equity

Table 6. Robustness test ruling out differences in loan quality

Read:

- Table 5 splits the marginal outcome test by the gender of the prior manager. Female workers face higher promotion standards under both male and female managers.
- Table 6 shows no significant evidence that marginally promoted female managers generate lower loan quality, measured by foreclosure outcomes.

Economic message: Bias is not concentrated only among male managers, which points toward broader stereotypes about women's managerial potential. The performance advantage of marginally promoted women is not offset by worse loan quality.

7.5 Table 7. Robustness test ruling out workforce-management differences and Table 8. Intermediate promotions and leaving the industry

Table 7
Robustness test ruling out gender differences in workforce management

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Change in team size		Fraction joining team		Fraction leaving team		Sales rank of leavers	
Subsample:	Male	Female	Male	Female	Male	Female	Male	Female
Promoted	−0.0179 (0.0132)	0.0019 (0.0024)	−0.0010 (0.0054)	0.0009 (0.0044)	−0.0034 (0.0056)	−0.0019 (0.0033)	−0.4038 (0.4724)	−0.2573 (0.3284)
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Observations	113,425	60,847	113,425	60,847	113,425	60,847	113,425	60,847
First-stage F-stat	48.8083	23.1582	95.7811	45.5624	95.7811	45.5624	95.7811	45.5624
p-value		.1400		.7884		.8206		.7991

This table reports robustness tests examining managers' workforce management practices. We use IV regressions to test for differences between the marginally promoted male versus female manager's impact on four aspects of workforce management. The unit of observation is at the loan officer-year level. The dependent variable in columns 1 and 2 is the average change in team size during a promoted manager's tenure; the dependent variable in columns 3 and 4 is the fraction of loan officers joining their team; the dependent variable in columns 5 and 6 is the fraction of loan officers leaving their team; and the dependent variable in columns 7 and 8 is the sales rank of the loan officers who leave the team. All the dependent variables are demeaned at the firm-year level and averaged over a manager's tenure. The key independent variable, *Promoted*, is an indicator that equals one if the loan officer is promoted in the next year and zero otherwise. We instrument for *Promoted* with a leave-out measure of the average promotion rate of loan officers at the same firm in the same year, excluding the focal loan officer and other officers working at the same branch. The sample is split on gender in order to identify differences between the marginally promoted male manager (columns 1, 3, 5, and 7) and the marginally promoted female manager (columns 2, 4, 6, and 8). The final row reports *p*-values from a Wald test of equality at the margin, that is, between the *Promoted* coefficients in the male/female tests. The controls include *Tenure*, *White*, and volume decile bins interacted with year. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7. Robustness test ruling out workforce-management differences

Table 8
Intermediate promotions and leaving the industry

A. Intermediate promotions

	(1)	(2)	(3)	(4)	(5)
	Total	Count		Percentage	
		Male	Female	Male	Female
Junior loan officer	56	49	7	87.50	12.50
Loan officer	591	450	141	76.14	23.86
Senior loan officer	282	231	51	81.91	18.09
Manager	178	141	37	79.21	20.79
Overall	1,107	871	236	78.68	21.32

B. Leaving the industry

	(1)	(2)	(3)	(4)
Dependent variable:	Leave the industry			
Female	0.0005 (0.0009)	0.0008 (0.0011)	−0.0021** (0.0011)	−0.0018* (0.0010)
Tenure			0.0094*** (0.0003)	0.0092*** (0.0003)
White			0.0015 (0.0012)	0.0018 (0.0012)
log(number of loans)			−0.0331*** (0.0006)	
Firm x year FE	Y	Y	Y	Y
Branch FE	N	Y	Y	Y
Volume decile x year FE	N	N	N	Y
Observations	174,015	171,182	171,182	171,182
R-squared	.0594	.1088	.1414	.1505

This table reports summary statistics on intermediate promotions and regression results on leaving the industry. Panel A provides summary statistics of job titles using a 5% random sample of loan officers matched to LinkedIn profiles when available, split by gender. The unit of observation is at the job spell level, and the various job titles from LinkedIn are consolidated into the four categories "Junior loan officer," "Loan officer," "Senior loan officer," and "Manager." Panel B reports OLS regressions testing whether loan officers' gender affects the likelihood of leaving the industry. The unit of observation is at the loan officer-year level. The dependent variable is an indicator for the loan officer leaving the industry in the next year. The key independent variable is the *Female* indicator. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8. Intermediate promotions and leaving the industry

Read:

- Table 7 examines team-size growth, recruiting, turnover, and sales rank of leavers; the differences between marginal male and female managers are not statistically meaningful.
- Table 8 shows that intermediate title patterns do not plausibly explain the main results, and women are not more likely to leave the industry after controls.

Economic message: The core bias interpretation survives several alternative explanations. The observed marginal female advantage is not simply because men outperform on hidden nonsales tasks, nor because women systematically leave the industry at higher rates.

7.6 Table 9. Manager gender and loan officers' performance and Table 10. Firms' gender promotion practices and firm-level outcomes

Table 9
Manager gender and loan officers' performance

A. Loan volume

	(1)	(2)	(3)	(4)
Dependent variable:	Log (number of loans)		Log (loan volume)	
<i>Female manager</i>	0.0039 (0.0198)	-0.0226 (0.0233)	0.0144 (0.0232)	-0.0190 (0.0274)
<i>Female manager x female LO</i>		0.0639** (0.0274)		0.0794** (0.0325)
LO tenure control	Y	Y	Y	Y
LO FE	Y	Y	Y	Y
Branch FE	Y	Y	Y	Y
Firm x year FE	Y	Y	Y	Y
Observations	144,254	144,254	144,254	144,254
R-squared	.7846	.7846	.7749	.7749

B. Loan quality

	Foreclosure ratio		Foreclosure amount	
<i>Female manager</i>	-0.0006 (0.0008)	0.0002 (0.0009)	0.0001 (0.0001)	0.0002 (0.0001)
<i>Female manager x female LO</i>		-0.0018 (0.0011)		-0.0002 (0.0002)
LO tenure control	Y	Y	Y	Y
LO FE	Y	Y	Y	Y
Branch FE	Y	Y	Y	Y
Firm x year FE	Y	Y	Y	Y
Observations	144,254	144,254	144,254	144,254
R-squared	.4825	.4825	.4244	.4244

This table reports OLS difference-in-differences regressions examining the average effect of having a female manager on subordinate loan officers' performance. The unit of observation is at the loan officer-year level. Panel A focuses on loan volume measures. The dependent variables in columns 1 and 2 and columns 3 and 4 are $\log(\text{number of loans})$ and $\log(\text{loan volume})$, respectively. Panel B focuses on loan quality measures. The dependent variables in columns 1 and 2 and columns 3 and 4 are *Foreclosure ratio* and *Foreclosure amount*, respectively. The key independent variables are *Female manager*, which indicates the loan officer's current manager is female, and its interaction with *Female LO*, which indicates the loan officer herself is female. Table A.1 in the appendix defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
Firms' gender promotion practices and firm-level outcomes

A. Loan volume

	(1)	(2)	(3)	(4)
Dependent variable:	log(number of loans)		log(loan volume)	
<i>Gender gap in promotions</i>	-18.9036*** (3.3903)	-5.5565*** (1.0969)	-19.8675*** (3.4551)	-5.4067*** (1.1304)
Firm FE	N	Y	N	Y
Year FE	Y	Y	Y	Y
Observations	2,314	2,204	2,314	2,204
R-squared	.0417	.9002	.0484	.8999

B. Loan quality

	Foreclosure ratio		Foreclosure amount	
<i>Gender gap in promotions</i>	-0.0189 (0.0230)	0.0305 (0.0196)	-0.0059 (0.0083)	-0.0009 (0.0091)
Firm FE	N	Y	N	Y
Year FE	Y	Y	Y	Y
Observations	2,314	2,204	2,314	2,204
R-squared	.1208	.4943	.0662	.4286

C. Firm employment and exits

	log(number of employees)		Firm exit next year	
<i>Gender gap in promotions</i>	-17.4994*** (2.7449)	-4.0497*** (0.8501)	1.0996*** (0.2298)	0.4877 (0.3129)
Firm FE	N	Y	N	Y
Year FE	Y	Y	Y	Y
Observations	2,314	2,204	1,907	1,820
R-squared	.0208	.9152	.0126	.3487

This table reports OLS regressions of firm outcomes on a firm-level measure of gender equity in promotions. The unit of observation is at the firm-year level. Panel A focuses on loan volume measures. The dependent variables in columns 1 and 2 and columns 3 and 4 are the firm's $\log(\text{number of loans})$ and $\log(\text{loan volume})$, respectively. Panel B focuses on loan quality measures. The dependent variables in columns 1 and 2 and columns 3 and 4 are the firm's *Foreclosure ratio* and *Foreclosure amount*, respectively. Panel C focuses on firm employment and exits. The dependent variables in columns 1 and 2 and columns 3 and 4 are the firm's $\log(\text{number of employees})$ and *Firm exit next year*, respectively. The key independent variable, *Gender gap in promotions*, is the difference between the male promotion rate and the female promotion rate at the firm that year. Table A.1 in the appendix defines all variables. The standard errors are clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9. Manager gender and loan officers' performance

Table 10. Firms' gender promotion practices and firm-level outcomes

Read:

- Table 9 shows that female loan officers become more productive under female managers, with a positive interaction between female manager and female loan officer for both number of loans and loan volume.
- Table 10 shows that firms with larger gender promotion gaps have lower loan volume and employment; loan quality does not appear to deteriorate at firms with smaller gaps.

Economic message: The paper's cost evidence is consistent with Becker's prediction that biased employment practices are costly. Promoting fewer women is associated with weaker sales, smaller employment, and weaker survival patterns.

7.7 Variable definitions: Table A.1. Variable definitions

Table A.1 Variable definitions	
Variables	Definition
<i>Promoted</i>	Equals one if the loan officer is promoted to manager and zero otherwise
<i>Externally promoted</i>	Equals one if the loan officer receives a promotion to manager in a different mortgage company and zero otherwise
<i>log(number of loans)</i>	Natural logarithm of the number of loans originated
<i>log(loan volume)</i>	Natural logarithm of the total dollar amount of loans originated
<i>Foreclosure ratio</i>	The ratio of number of foreclosures over number of loans originated
<i>Foreclosure amount</i>	The ratio of total dollar amount of foreclosures over total dollar amount of loans originated
<i>Female</i>	Equals one if the loan officer is female and zero otherwise
<i>White</i>	Equals one if the loan officer is white, and zero otherwise
<i>Tenure</i>	Number of years working at the firm
<i>Change in team size</i>	Change in the number of loan officers in the team each year, averaged over each manager's tenure
<i>Fraction joining team</i>	Fraction of the team that is new each year, averaged over each manager's tenure
<i>Fraction leaving team</i>	Fraction of the team that is leaving each year, averaged over each manager's tenure
<i>Sales rank of leavers</i>	Average sales percentiles of loan officers who leave the manager's team
<i>log(number of employees)</i>	Natural logarithm of the total number of employees
<i>Firm exit next year</i>	Equals one if the firm exits the data in the next year and zero otherwise
<i>Gender gap in promotions</i>	The difference between the male promotion rate and the female promotion rate at the firm

Table A.1. Variable definitions

Read:

- Appendix Table A.1 defines the key outcome and explanatory variables used throughout the empirical analysis.
- The definitions clarify the distinction between worker-level variables, manager value-added variables, and firm-level gender-equity measures.

Economic message: The table is useful for reading the paper because the same variables connect three layers of evidence: worker promotion gaps, marginal outcome tests, and firm-level cost regressions.

8 Short summary

Huang, Mayer, and Miller study gender bias in promotions at U.S. mortgage companies. Women are less likely to become managers than men with similar measured performance and experience. Because standard promotion regressions cannot prove bias, the paper applies Becker’s outcome-test logic at the promotion margin. Using a leave-out firm promotion rate instrument, the authors show that marginally promoted female managers outperform marginally promoted male managers, implying that firms require higher managerial ability from women before promotion.

The paper also tests possible mechanisms and costs. Bias is present under both male and female prior managers, suggesting that broad stereotypes, not only male animus, are important. The authors rule out key alternative explanations involving loan quality, workforce management, intermediate promotions, and differential industry exit. Finally, firms with larger gender promotion gaps perform worse on loan volume and employment outcomes, consistent with biased employment practices being costly.

9 Conclusion

Core conclusion. The paper provides direct evidence that gender bias affects promotions at financial institutions. Female loan officers face lower promotion rates and, when promoted at the margin, outperform comparable male managers.

Economic intuition. If firms promote men at a lower threshold, then some lower-value-added male managers replace female workers with higher managerial potential. This reduces female career advancement and can also reduce firm output.

Incremental contribution revisited. The paper moves the literature from documenting gender gaps to testing gender bias directly with a marginal outcome test. It also links bias to firm costs, thereby connecting labor-market discrimination to corporate performance.

Implications. The findings suggest that simply increasing female representation among evaluators may not eliminate bias if inaccurate stereotypes are widely held. Better promotion systems may need explicit checks on promotion thresholds, manager evaluations, and objective ex post performance.

Limitations. The firm-performance results are mostly correlational, and the setting is mortgage companies rather than all financial firms. Still, the worker-level outcome tests are strong evidence for bias in the specific promotion margin studied.

Takeaway. The paper’s main message is that the early-career “broken rung” in finance is not only a disparity; it is consistent with a higher promotion bar for women and with real costs to firms.